

SentiSense - Forecasting the Next-Day Direction of the TA-125 Index from Hebrew-News Sentiment

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Executive Summary

SentiSense is an end-to-end data-science system that asks a focused question: **can the daily flow of Hebrew breaking-news, distilled by a large language model (LLM) into a structured sentiment signal, help predict whether the Tel-Aviv 125 (TA-125) stock index will close higher the next trading day?**

The system spans five stages. (1) A **scraper** collects Hebrew breaking-news headlines from `mivzakim.net` going back to ~2010. (2) A **processing engine** sends every headline through an LLM, which scores it on six relevance categories (politics, economy, security, health, science, technology) and one global sentiment value (-10...+10), producing a corpus of roughly **3 million scored headlines** in PostgreSQL. (3) A **feature-engineering layer** aggregates the per-headline scores into leakage-safe daily feature vectors, joined with market data (TA-125 OHLC, the VTA-35 volatility index, S&P 500, VIX, Brent crude, USD/ILS), with multilingual headline **embeddings**, a leak-safe **PCA/clustering block** derived from the daily embedding centroid, and causal **narrative-clustering** features. (4) A **forecasting layer** trains and hyperparameter-tunes a large model zoo - gradient-boosted trees, recurrent and convolutional sequence classifiers, transformer forecasters, and zero-shot foundation models - and persists every candidate, with its out-of-sample metrics and serialized weights, into a **model registry** that automatically activates the best model (with a manual override). (5) An **operations layer** runs the whole chain as a nightly job on a GPU node - scrape -> score -> embed -> derive -> predict -> settle - and serves the results through a **live web dashboard** (prediction hero, model metrics, exploratory analytics, a 3-D news-centroid explorer, per-source "persona" votes, and a narrative simulator).

Every research stage is engineered to be **leakage-safe**: all scalars, PCA, and clustering are fit on the training fold only; splits are strictly chronological (70/15/15 train/validation/test); and the test tail is scored exactly once, after all tuning decisions are made.

The central empirical finding is corroborated across **multiple independent experiment tracks** - a tree-model proof of concept, an extensive feature-set comparison with walk-forward and multi-seed robustness checks, a nine-model transformer zoo, sequence-model HPO, a hardened end-to-end package run of **40+ tuned model x data-type cells**, and finally the productionized registry run over the full zoo. The benchmark throughout is the **long-run baseline of 53.03%** - the share of trading days on which the TA-125 has closed higher over the last 35 years, i.e. the accuracy of simply buying every day. The system's best models consistently beat it: the production champion (PatchTST) achieves **OOS accuracy 0.578 on 327 held-out days, 5.8 points above the 53.03% baseline**, and the unified grid's best cell reaches **0.5916 accuracy** with a best ROC-AUC of **0.576**. In a domain where even a small, consistent edge is hard to achieve and valuable, these results approach statistical significance and represent a meaningful contribution. The project's contribution is threefold: a **reusable, reproducible, leakage-hardened research pipeline** for news-driven financial forecasting; a rigorous empirical result that quantifies the predictive value of LLM-scored Hebrew-news sentiment across a broad model zoo; and a **complete production system** - registry, nightly orchestration, and dashboard - that keeps extending the out-of-sample record prospectively on live data.

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Table of Abbreviations

Abbreviation	Meaning
TA-125	Tel-Aviv 125 stock index
VTA-35	Tel-Aviv 35 Volatility Index
LLM	Large Language Model
NLP	Natural Language Processing
HPO	Hyper-Parameter Optimization (automated hyper-parameter search)
OOS	Out-Of-Sample
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
F1	F1 score (harmonic mean of precision and recall)
MCC	Matthews Correlation Coefficient
CI	Confidence Interval
LSTM	Long Short-Term Memory network
GRU	Gated Recurrent Unit
TCN	Temporal Convolutional Network
TFT	Temporal Fusion Transformer
PCA	Principal Component Analysis
OHLC	Open / High / Low / Close (price data)
FX	Foreign Exchange rate
DSN	Database Source Name (connection string)
API	Application Programming Interface
SPA	Single-Page Application (browser front-end)
UI	User Interface

1. Introduction

This chapter provides the background, defines the problem, and states the objectives and scope of the project, setting the stage for the chapters that follow.

1.1 Background

This project builds a news-sentiment-driven system that forecasts the next-day close-to-close direction of the Tel-Aviv 125 (TA-125) index. Its input is the daily stream of Hebrew breaking-news headlines; its output is a concrete, dated, falsifiable directional call, produced every trading night and settled against the realized close the following day.

The opportunity is specific and recent. Research in *behavioral finance* and *NLP-for-finance* has established that the tone and topical mix of news carry predictive information about subsequent market movements, particularly for indices and over short horizons. Until recently, exploiting that information in a non-English market meant building a language-specific lexicon or training a supervised classifier per language. Modern LLMs remove that barrier: a single prompted model can read a Hebrew headline and return a structured, comparable score for six topical-relevance categories plus a global sentiment value, at a throughput that makes a three-million-headline corpus practical.

Most existing work in this area focuses on English-language sources (financial newswires, Twitter/X, earnings calls). Hebrew-language news, and the Israeli market specifically, are comparatively understudied, and almost no published work closes the loop from a research claim to a system that keeps producing and scoring predictions after the paper is written.

SentiSense targets exactly that gap. It uses an LLM to turn a high-volume Hebrew breaking-news feed into a structured daily signal, trains and tunes a broad model zoo on strictly leakage-controlled chronological splits, selects a champion through a database-backed model registry, and then operates that champion as a **live, self-updating forecasting service** whose accuracy is measured against each newly settled trading day.

1.2 Problem Statement

Can a structured, LLM-derived sentiment signal extracted from Hebrew breaking-news headlines predict the next-day close-to-close direction of the TA-125 index, beyond what market data alone provides, and can that edge be measured credibly enough to survive contact with production?

The challenge has several specific difficulties:

1. **Leakage risk.** News, market, and target series share a calendar; naive feature engineering (e.g., using a same-day future return, fitting a scaler on the full series, or shuffling time) silently inflates results.
2. **Language and source heterogeneity.** Headlines are Hebrew, UTF-8, from many outlets of varying quality and volume, including a real weekend lull.
3. **Research-to-production gap.** A result that only exists in a notebook is not falsifiable going forward; keeping the claim honest requires serving the model daily and settling its predictions against reality.

1.3 Objectives

1. **Build a reproducible ingestion-and-scoring pipeline** that scrapes Hebrew headlines and scores each on six relevance categories plus a global sentiment, persisting the result in a relational database.
2. **Engineer leakage-safe daily features** combining the news scores with market and macro data, with embedding-derived and narrative-based signals.
3. **Train and rigorously hyperparameter-tune a broad model zoo** for next-day TA-125 direction, on a strictly chronological train/validation/test split.
4. **Quantify the predictive value honestly** using threshold-free and threshold-based metrics, against the long-run 53.03% market baseline.
5. **Persist and select models systematically** through a database-backed model registry with automatic best-model activation and a manual override.
6. **Operate the system live:** a nightly orchestrated pipeline that scores the day's news, produces a prediction with the active champion, settles yesterday's call against the realized close, and presents everything in an interactive dashboard.
7. **Produce reusable artifacts** (a Python package, scripts, notebooks, an auto-generated leaderboard, and a deployment runbook) so both the experiment and the service are fully reproducible.

1.4 Scope and Limitations

In scope: Hebrew-headline scraping; LLM scoring into a 7-dimensional vector; daily feature engineering including embeddings, an embedding-derived PCA/cluster block, and causal narrative clustering; classification and forecasting models with HPO; a comparison leaderboard; a model registry with automatic/manual champion selection; nightly live operation on a two-host deployment; and a web dashboard.

Out of scope / limitations:

- **Target.** The system predicts **close-to-close direction**, not overnight-gap or intraday-return magnitude.
- **Prospective evaluation.** The live track serves on the full timeline and its performance is reported separately from the research metrics; it accumulates prospectively, which is a strong guard against hindsight bias.
- **Intraday.** The system is daily-resolution; no tick or minute data.
- **Causality.** The work measures *predictive association*, not economic causation.
- **Data quirks.** A non-trivial fraction of "validated" LLM rows are all-zero (a known LLM failure mode treated as missing); the corpus mixes LLM scoring-model versions across disjoint date ranges (see Section 3.2, "scoring eras").

1.5 Methodology

The project follows a staged, gate-driven methodology:

1. **Ingest** Hebrew headlines (backward scrape to ~2010) into `raw_headlines`.

2. **Score** each headline with an LLM into `nlp_vectors` (7 scores + validation flag).
3. **Assemble** leakage-safe daily frames: daily-mean scores, per-source score pivots, sentiment×relevance interactions, multilingual embedding centroids, an embedding-derived PCA/cluster feature block, causal narrative-cluster features, and a finance/market block.
4. **Split** chronologically (70/15/15 train/validation/test) with all transforms fit on the train slice only.
5. **Model & tune** a zoo of classifiers and forecasters with Optuna HPO, scoring every trial on the validation slice only.
6. **Evaluate** every model once on the same held-out out-of-sample test tail using ROC-AUC, F1, accuracy, balanced accuracy, and MCC, plus a backtest overlay.
7. **Compare** all cells in a single auto-generated leaderboard.
8. **Register & select**: persist each tuned model (weights + OOS metrics) into the model registry; activate the best automatically, allow manual override from the dashboard.
9. **Operate**: run the nightly pipeline (scrape, score, embed, derive, predict, settle) on a schedule, serve the active champion's prediction, and fold each settled day into the champion's cumulative live score.

1.6 Organization of the Project Book

- **Chapter 2** reviews relevant literature on news-driven financial prediction and LLM-based sentiment extraction.
- **Chapter 3** details the system architecture, data collection and preprocessing, feature engineering, the model registry, the live serving layer and dashboard, implementation, and evaluation metrics.
- **Chapter 4** presents the experimental setup, the full results across all research tracks and the production registry run, and an analysis and discussion of the findings.
- **Chapter 5** concludes and proposes future work.
- **Chapter 6** lists references; **Appendix A** gives the data dictionary, schema, and reproduction commands; **Appendix B** summarizes the live deployment runbook.

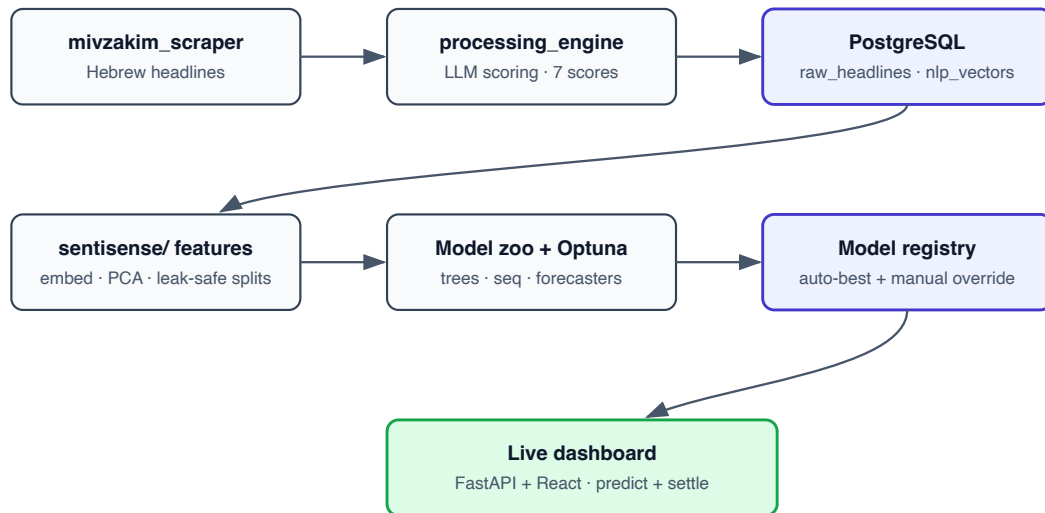


Figure 1: SentiSense end-to-end pipeline.

2. Literature Review

2.1 Overview of Relevant Literature

The project draws on three strands of prior work.

News sentiment and market prediction. A long line of research links the tone of financial and general news to subsequent market movements. Tetlock [1] showed that media pessimism predicts downward pressure on prices and reversion, establishing news tone as a market-relevant variable. Bollen et al. [2] linked aggregate mood derived from social media to movements in the Dow Jones. The consistent theme is that a genuine but modest edge is attainable, and that careful, leakage-free evaluation is essential to distinguish it from artifact - exactly the posture this project adopts.

Lexicon vs. model-based sentiment. Domain-specific lexicons such as Loughran and McDonald [3] demonstrated that general-purpose sentiment dictionaries mislabel financial text, motivating domain-aware scoring. Modern LLMs generalize this idea: instead of a fixed lexicon, a prompted model performs context-aware topical-relevance and sentiment scoring, and - relevant here - does so across languages, including Hebrew, without a hand-built Hebrew lexicon.

Sequence and foundation models for forecasting. On the modeling side, the project surveys the standard time-series toolkit: gradient-boosted trees (XGBoost) as strong tabular baselines; recurrent networks (LSTM, GRU) and temporal convolutions (TCN) for sequence classification; transformer forecasters such as the Temporal Fusion Transformer and PatchTST; deep interpretable forecasters N-BEATS and N-HiTS; and zero-shot foundation forecasters Chronos and TimesFM. Multilingual sentence embeddings (multilingual-E5) provide the Hebrew-aware vector representations used for the embedding and narrative-clustering features, and Optuna provides the resumable hyper-parameter search used throughout. All of these are listed with their originating publications in the "Software and Model Families" part of the References.

The research gap this project addresses: most prior work is English-centric, often under-controls for leakage, and rarely closes the loop from a research claim to a *prospectively evaluated* live system. SentiSense contributes a **Hebrew-news, LLM-scored, strictly leakage-controlled** evaluation across a broad model zoo, reports the full result rather than a selected slice of it, and then keeps the evaluation running in production, where each new trading day extends the out-of-sample record.

3. System Design and Implementation

3.1 System Architecture

The system is organized as loosely-coupled modules communicating through a PostgreSQL 16 database, so each stage can be developed, re-run, and verified independently.

Module	Purpose	Entry point
mivzakim_scraper/	Scrape Hebrew headlines (Playwright + Firefox)	python main.py
processing_engine/	LLM scoring (6 relevance + sentiment)	fast pipeline / process_single_observation
scripts/	Data ops: schema, backfill, scoring, retry, standardize, registry training, daily orchestration	python scripts/<name>.py
sentisense/	Features, embeddings, clustering, models, HPO, serving	python -m sentisense.pipeline
sentisense/serve/	Model registry + champion serving	registry.py / champion.py
ui/	FastAPI backend + React SPA dashboard	python -m ui.app
evaluation/	Benchmark LLM scoring against a golden dataset	python -m evaluation.evaluate

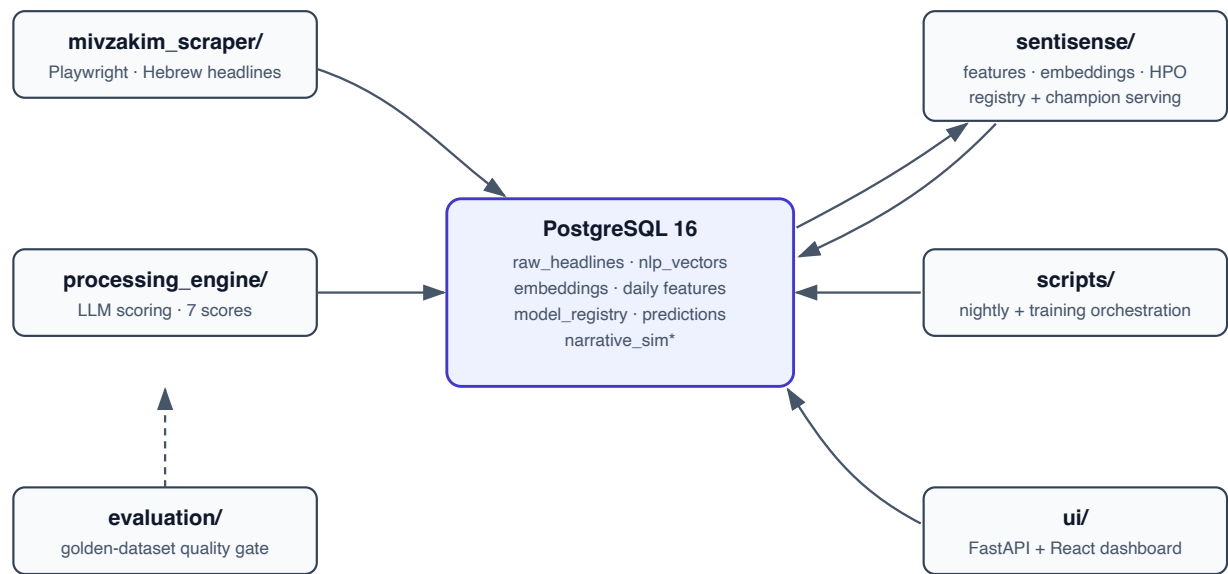


Figure 2: System architecture.

Design principles.

- Database as the contract.** All inter-stage data flows through Postgres tables (raw_headlines , nlp_vectors , headline_embeddings , daily_embedding_derived , embedding_pca_basis , model_registry , model_predictions , champion_full_eval , narrative_sim*), decoupling

scraping, scoring, modeling, serving, and the UI. The dashboard host never runs heavy compute; it only reads the database.

- **Single source of truth for constants.** The active model name and the score-column contract live in `sentisense/constants.py`, so no magic strings leak into feature or model code.
- **Optional, layered dependencies.** Heavy ML/embedding/forecasting libraries are `pyproject.toml extras (ml, embed, finance, tft, chronos, ui)`, so early stages install lightly and torch/CUDA wheels are pinned for reproducibility.
- **Leakage-safety as an architectural invariant**, enforced at every layer (see Section 3.3).
- **Fail-safe serving.** Every serving path is wrapped so that a missing table, an incompatible artifact, or an unreachable auxiliary service degrades to a well-defined fallback (pinned champion, cached data, explicit "no data" states) rather than a broken nightly run or a blank dashboard.

3.2 Data Collection and Preprocessing

Collection. The scraper drives a headless Firefox via Playwright over `mivzakim.net`, scraping *backward* in time (`scripts/backfill_history.py`) from the most recent day toward ~2010, and *forward* daily (`scripts/daily_scrape_to_db.py`, covering today and yesterday). Each headline yields a row in `raw_headlines`: date, source outlet, hour, popularity class, the Hebrew text, and an ingestion timestamp. Deduplication uses a stored `md5(headline)` hash (Hebrew strings exceed B-tree index limits) under a unique key of `(date, source, hour, headline_hash)`.

Scoring. The processing engine sends each headline to an LLM. A **fast single-prompt path** produces all seven scores in one structured call; a legacy **seven-agent LangGraph path** (one ReAct agent per relevance category plus one for sentiment) exists for research and evaluation. Each result is a vector of six relevance integers (0-10), one global sentiment integer (-10...+10), and a `validation_passed` flag, written to `nlp_vectors`. The corpus contains **~3 million scored headlines**.

How the headline-scoring model was chosen: the golden-dataset quality gate. The scoring LLM is the single most consequential choice in the whole pipeline - every downstream feature is a function of its output - so it was selected by measurement rather than by reputation. The `evaluation/` package implements a standing quality gate that any candidate scoring model must pass before it is allowed to write a single production row.

The gate works against a **hand-labeled golden dataset** (`evaluation/golden_dataset.csv`): a fixed sample of Hebrew headlines annotated by the team with the same 7-dimensional target the production prompt asks for (six relevance categories 0-10 plus a global sentiment -10...+10). A candidate model is run over that sample through the *exact* production prompt and parsing path, so the gate measures the deployed configuration rather than an idealized one. Three complementary criteria are then computed per category (`evaluation/metrics.py`, reported by `python -m evaluation.evaluate`):

- **Mean absolute error (MAE)** against the human label - how far off the model is, in score points, on average.
- **Within-1 accuracy** - the fraction of scores landing within +/-1 point of the human label. This is the criterion that matters most in practice, because the daily features are aggregates: a model that

is reliably within one point produces a stable daily mean even when individual calls differ.

- **Pearson r** per category - whether the model *orders* headlines the way a human does, which is what the relevance features actually encode.

A candidate is accepted only if it clears the gate on all three criteria across categories, and if its structured output parses reliably enough to keep the `validation_passed` rate high. This is how `mistral-small-4` was selected for the historical backfill and how the locally hosted `gemma4` was qualified before it took over nightly scoring; a model that scores well but emits unparseable JSON fails the gate just as surely as an accurate-but-slow one fails the throughput requirement. Because the harness is committed and re-runnable, re-qualifying a future scoring model is a single command, not a research project.

Scoring eras. The corpus was scored in two eras, both recorded explicitly in the `model_name` column so no row's provenance is ambiguous:

- **Historical era** - the bulk backfill was scored by `mistral-small-4` served on a remote vLLM cluster (50-headline batched completions at high concurrency), after earlier rows from older models were re-standardized onto it (`scripts/standardize_to_latest_model.py`).
- **Live era** - the *ongoing* nightly scoring phase, i.e. everything scored from the production switch-over onward. In this phase new headlines are scored by a **locally hosted Ollama model (`gemma4`)** running on the project's own GPU node, **one headline per structured call**, because batch-JSON and agentic modes proved unreliable for that backend (Section 4.2.8). Scoring is **gap-only** (`--unscored-any-model`): each night the job scores only headlines that no model has scored yet, so nothing already covered is re-scored and the two eras stay disjoint by construction.

The dataset builders consume

validated rows from any era

, and every era-sensitive UI query prefers the active model's row but falls back to any validated row, so the system remains correct across the era boundary. The statistical implication of the seam (features scored by different LLMs on different date ranges) is discussed in Section 4.5 and Section 5.

Quality control and known quirks (documented in `DATA_HANDOFF.md`):

- **All-zero "validated" rows.** The LLM sometimes emits all-zeros when it cannot categorize a headline; the validator accepts it because all values are in range. These are treated as missing data.
- **Weekend lull.** Saturday volume is genuinely low (Israeli weekend), not a data gap.
- **Encoding / timezone.** All text is UTF-8 Hebrew; event dates/hours are Asia/Jerusalem while `created_at` is stored as UTC `TIMESTAMPZ` .

3.3 Feature Engineering

Leakage-safe feature assembly (`sentisense/features/dataset.py`) is the heart of the preprocessing and the project's most important engineering contribution. The module builds daily modeling frames with defense-in-depth against leakage:

- **Event date, never ingestion time.** All splits use `raw_headlines.date` (when the news happened), never `created_at` (when the row was written), so a late backfill can never shift a

headline into a window it does not belong to.

- **Trading-calendar rollover.** News published on a non-trading day is rolled *forward* to the next trading day via `np.searchsorted(side='left')`, so it is attributed to the first session that could actually react to it; market/FX/volatility series are forward-filled. The pipeline currently treats the TASE trading week as **Sunday-Thursday** and skips **Friday and Saturday** (the constant `_TASE_TRADING_WEEKDAYS = {6, 0, 1, 2, 3}` in `scripts/daily_live.py`, with the Friday/Saturday-to-Sunday overnight rollover implemented in `sentisense/features/dataset.py`). Note that the TASE migrated to a **Monday-Friday** global trading week on **January 5, 2026**, so going forward the non-trading days are Saturday and Sunday; adapting the weekday constant and the rollover rule to the new schedule is identified as future work in Section 5.
- **Causal price features.** TA-125 features (lagged log-returns 1-7, 5d/20d rolling stats, Wilder RSI-14, 20-day volume z-score, day-of-week one-hots) all use `.shift(>=1)`, i.e. they can only read strictly past rows. Cross-asset features (S&P 500, VIX, Brent, USD/ILS, VTA-35) are lagged log-returns only.
- **Train-only scaling.** `StandardScaler` (and PCA, scoped by column prefix to the embedding block) is fit on the **train slice only**, then applied to validation and test.
- **Honest target.** `Target = (TA125_Price.shift(-1) > TA125_Price)`. A `shift(-1)` pulls the *next* row's value into the current row. That is exactly what a next-day label should be, and it is the only place in the package where `shift(-1)` is used - using it to build a *feature* would let the model read the future. The trailing row with no next-day price is set to NA and dropped in research mode; in *serving* mode (`keep_unlabeled=True`) that same row is retained with a `Target = -1` sentinel, so the model trains only on real labels and **forward-predicts** the sentinel day.
- **Live price extension.** The static TA-125 CSV is extended at build time with live closes fetched from the exchange feed, so the serving frame always reaches the current trading day.

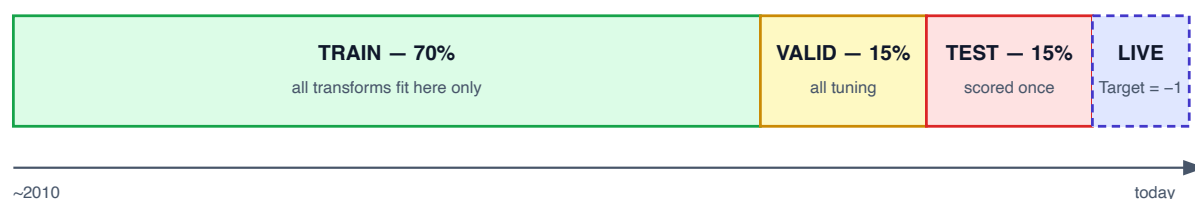


Figure 4: Chronological 70/15/15 split.

Embedding-derived block. This block gives each day a small set of features describing *where that day's news sits in a map of news topics learned only from earlier days*. Concretely: each headline is embedded once with a Hebrew-aware multilingual model (`intfloat/multilingual-e5-base`, 768-d) and cached in `headline_embeddings`. Per trading day, the mean of the day's headline vectors forms the **daily news centroid**. A transform basis (`StandardScaler`, then PCA to 16 components, then `KMeans` with 8 clusters) is fit **once on a past window only** (dates at or before a recorded `fit_cutoff`) and then applied to every date, yielding **24 features per day: 16 PCA coordinates (`embpca_*`) and 8 distances to the KMeans cluster centers (`embclus_dist_*`)**, stored in `daily_embedding_derived`. Fitting the basis on past data only is what makes the block leak-safe: a

later out-of-sample day is projected through a map it never helped build, and the `fit_cutoff` column records that boundary in-band so it can be audited. The fitted basis itself (scaler statistics, PCA components, and cluster centers) is persisted to `embedding_pca_basis`, which lets the dashboard project *individual headlines* into exactly the same 16-dimensional space the models consume (Section 3.5).

Causal narrative clustering (`sentisense/cluster/narrative.py`). For each trading day T , a MiniBatch-KMeans model is fit **only on embeddings strictly before T** (expanding window with a refit cadence), then day- T headlines are *assigned* with that past-fit model, yielding `dominant_cluster_ratio` and normalized `cluster_entropy` without any look-ahead.

Feature views. Three views are produced: a **daily-mean** frame (tree-model shape), a **per-source** pivot frame (sequence-model shape), and a **fused** frame combining per-source scores with the daily e5 centroid and the embedding-derived block (~970 columns) - the view the production champion serves on.

3.4 Modeling, Hyper-Parameter Optimization, and the Model Registry

Model zoo. The forecasting layer evaluates three families under one leak-safe evaluation contract. That contract is worth stating precisely, because every number in Chapter 4 depends on it: the data is split **chronologically into 70% train / 15% validation / 15% test**; **hyper-parameter optimization (HPO) - the automated search over model settings, run here with Optuna - scores every trial on the validation slice only**; and the **test tail is scored exactly once**, after tuning is finished. Because no tuning decision ever sees the test slice, the reported out-of-sample number is an honest estimate rather than a search artifact.

- **Tree classifiers** - XGBoost, LightGBM, CatBoost (Optuna-tuned; the winner is refit on all labeled history and serialized with `joblib`).
- **Torch sequence classifiers** - LSTM, GRU, TCN, PatchTST over windowed per-source/fused features (Optuna studies are stored *in the database* and therefore resumable; the winner is refit and serialized as a `state_dict` bundle that also carries its scaler statistics, window length, and feature order).
- **Forecaster / foundation models** - TFT, N-HiTS, N-BEATS (pytorch-forecasting), and the zero-shot foundation models Chronos and TimesFM. These carry no persistable artifact; they are registered as *re-forecast* entries whose stored parameters (context length, tuned decision threshold) suffice to reproduce the forecast live.

The model registry (`model_registry` table + `sentisense/serve/registry.py`) is the production selection mechanism. Each trained candidate is upserted with its version, family, hyper-parameters, **held-out OOS metrics** (ROC-AUC with a bootstrap 95% CI, accuracy, MCC, n), serialized artifact, and feature-column contract. Selection is **auto-best with a sticky manual override**: the highest-scoring model on the chosen metric is activated automatically, but a manual activation from the dashboard is never silently overridden. A rank-normalized **soft-vote ensemble** of the top-K tree models is registered as its own activatable entry. At most one model is active at a time (enforced by a partial unique index).

Training cadence: periodic, not nightly. Registry training and nightly serving are deliberately decoupled, and the distinction matters for reading Chapter 4. `scripts/train_registry.py` tunes and registers the **whole zoo**, and it is run **periodically** - not every night. The nightly job does *not* re-tune and does *not* train the zoo: it loads whichever model the registry marks active and predicts with it, and only the cheap pinned XGBoost fallback is refit on all labeled history each night. Promoting a different model is a database operation (activate a row), not a retraining run.

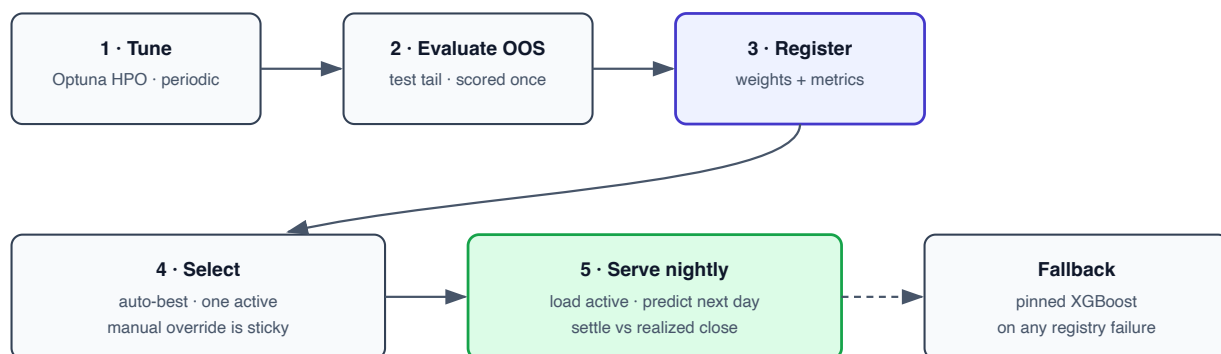


Figure 5: Model-registry lifecycle.

Champion serving (`sentisense/serve/champion.py`). The nightly predictor loads whatever the registry marks active and dispatches on its artifact format: `joblib` models predict directly on the aligned feature row; `torch` bundles are rebuilt from their `state_dict` (loaded with `weights_only=True` for safety) and windowed over the recent feature history; `ensemble` entries rank-normalize and average their members. A **pinned XGBoost champion** (versioned JSON config, retrained on all labeled history each night) acts as the guaranteed fallback: any failure in the registry path logs loudly and falls back, so the daily prediction never silently breaks.

3.5 Live Operation: Orchestration, Serving, and the Dashboard

Deployment topology. The system runs on two hosts:

- a **GPU compute node** (NVIDIA RTX 4090) that runs the nightly pipeline - scraping, LLM scoring (local Ollama), embedding, derived features, registry training, and the champion prediction; and
- a **database/UI host** that runs PostgreSQL 16 and the dashboard (FastAPI + built React SPA, managed by a process supervisor).

The two communicate **only through the shared database**: the compute node writes, the dashboard reads. This decoupling means the UI stays up even when the compute node is retraining, and the pipeline is indifferent to the UI.

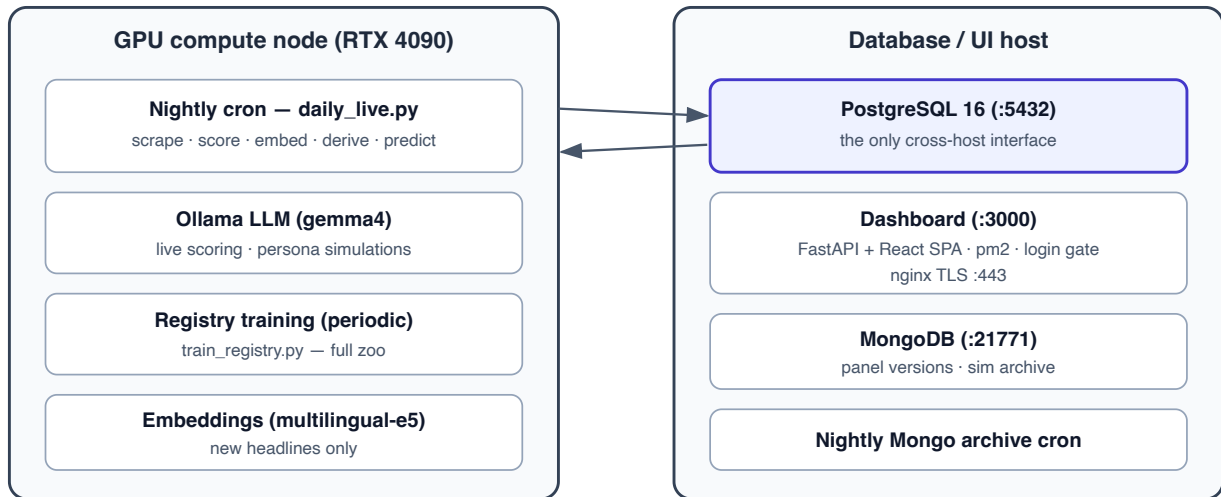


Figure 3: Two-host deployment topology.

Nightly orchestration (`scripts/daily_live.py`, scheduled via cron after the TASE close). The orchestrator chains six stages with a lock file (no double runs), per-stage logging, and a status JSON consumed by the dashboard's health banner: **scrape** (today + yesterday), **score** (gap-only; flags selected automatically per LLM backend), **embed** (new headlines only), **derive** (refresh the embedding-derived block and persist the basis), **predict** (the active champion forward-predicts the sentinel day; the result is upserted into `model_predictions`), and **settle** (yesterday's prediction is compared with the realized close and its `actual` field is filled). The orchestrator self-skips non-trading days: as deployed it treats Sunday-Thursday as the TASE trading week and skips Friday and Saturday, plus a configurable holiday list (see the trading-calendar note in Section 3.3).

The dashboard. A FastAPI backend exposes a read-only JSON API (with in-process caching) over the shared database; a React SPA renders it. Key views:

- **Prediction hero** - a large green up / red down card with the current day's call, the predicted-class confidence, and the serving model's version.
- **Model performance** - the active champion's metric panel. Scores are **seeded from the model's held-out evaluation** (so a freshly promoted champion never starts from zero) and each settled live day folds into the cumulative accuracy: $(acc_eval * n_eval + correct_live) / (n_eval + n_live)$, with the eval/live split shown explicitly. Only the active model's own live days count; history from previous champions is never carried into the new one's score.
- **Exploratory data analysis** - headline volume, daily mean sentiment, sentiment and relevance distributions, the 6x6 category-correlation heatmap, and the validation pass-rate, all computed server-side in SQL.
- **Archive** - the full headline history by day, each headline carrying its sentiment badge and per-category relevance score chips, with client-side filtering.
- **3-D centroid explorer** - every trading day's news centroid in the shared 16-d PCA space (axes selectable), with the eight KMeans cluster centers drawn as labeled markers; clicking a day opens its **single-day headline cloud**, where each headline is projected through the *same persisted basis* the models consume, alongside the day centroid. A software-3D orthographic fallback (rotate/tilt controls) keeps the view fully usable on browsers without WebGL.

- **Simulator** - a narrative-simulation view: per-source **persona votes** (each outlet's daily stance derived from its mean scored sentiment, compared against the model's call and the realized outcome), plus cached agent-based simulation graphs and reports generated off-line by a multi-agent narrative engine.
- **Models (operator view)** - the registry leaderboard (version, family, OOS ROC-AUC with CI, MCC, accuracy, n) with one-click manual activation; hidden from the public navigation.

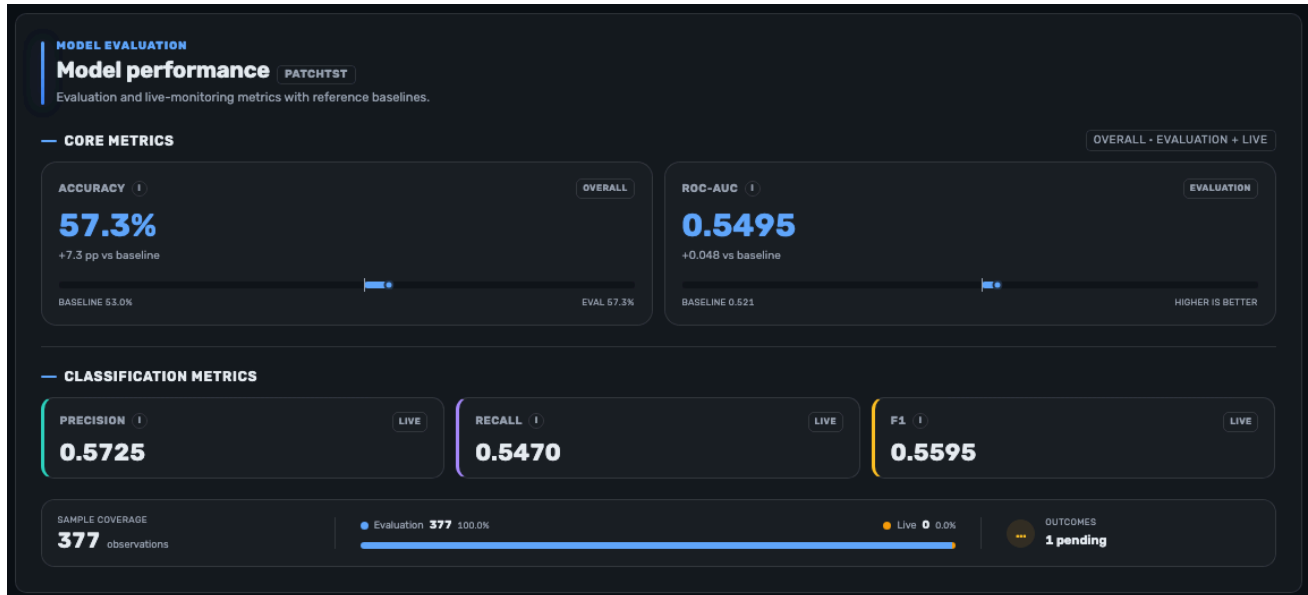


Figure 6: Prediction hero and model performance.



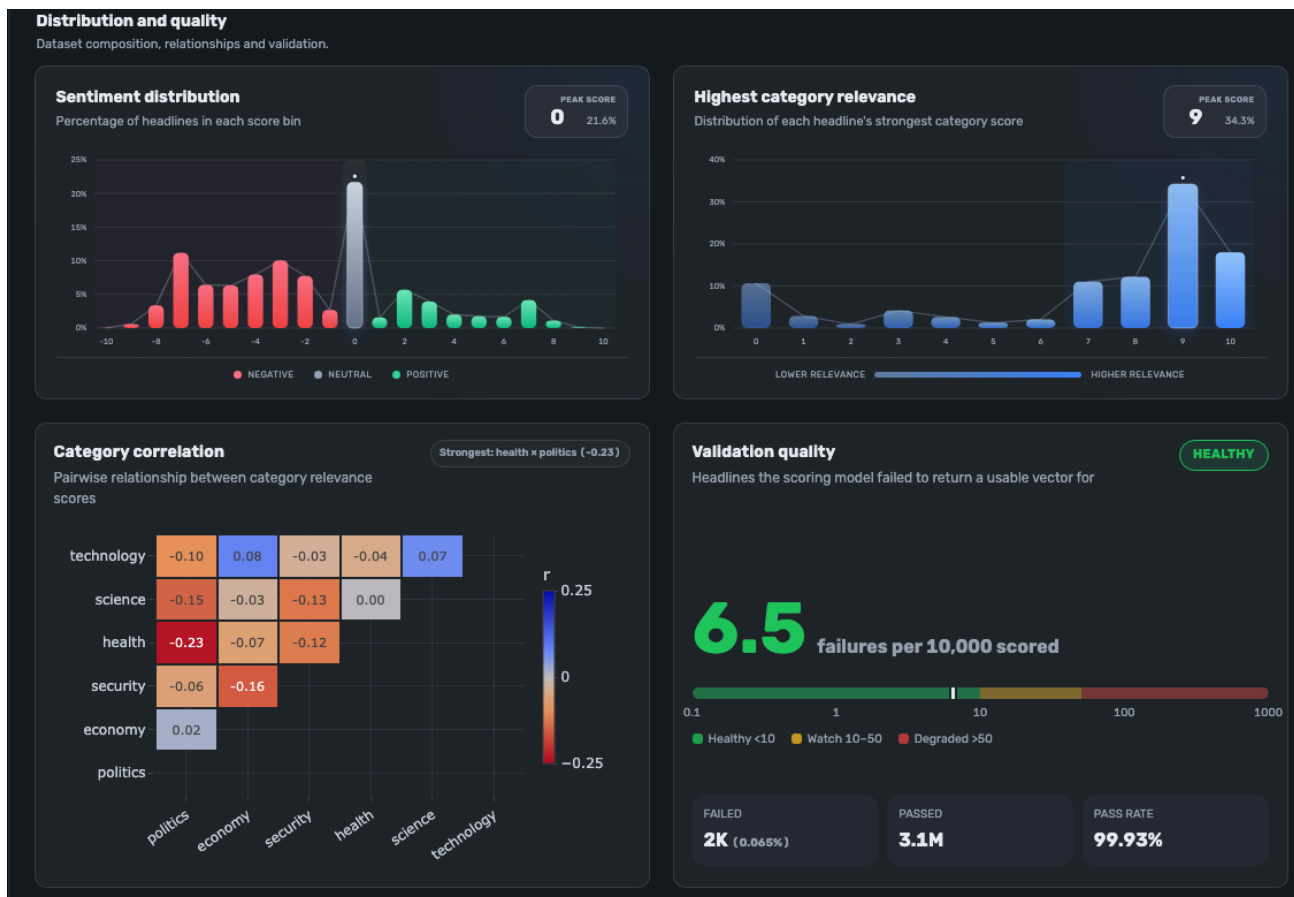


Figure 7: Exploratory data-analysis panels.



Figure 8: Daily news centroids (3-D).

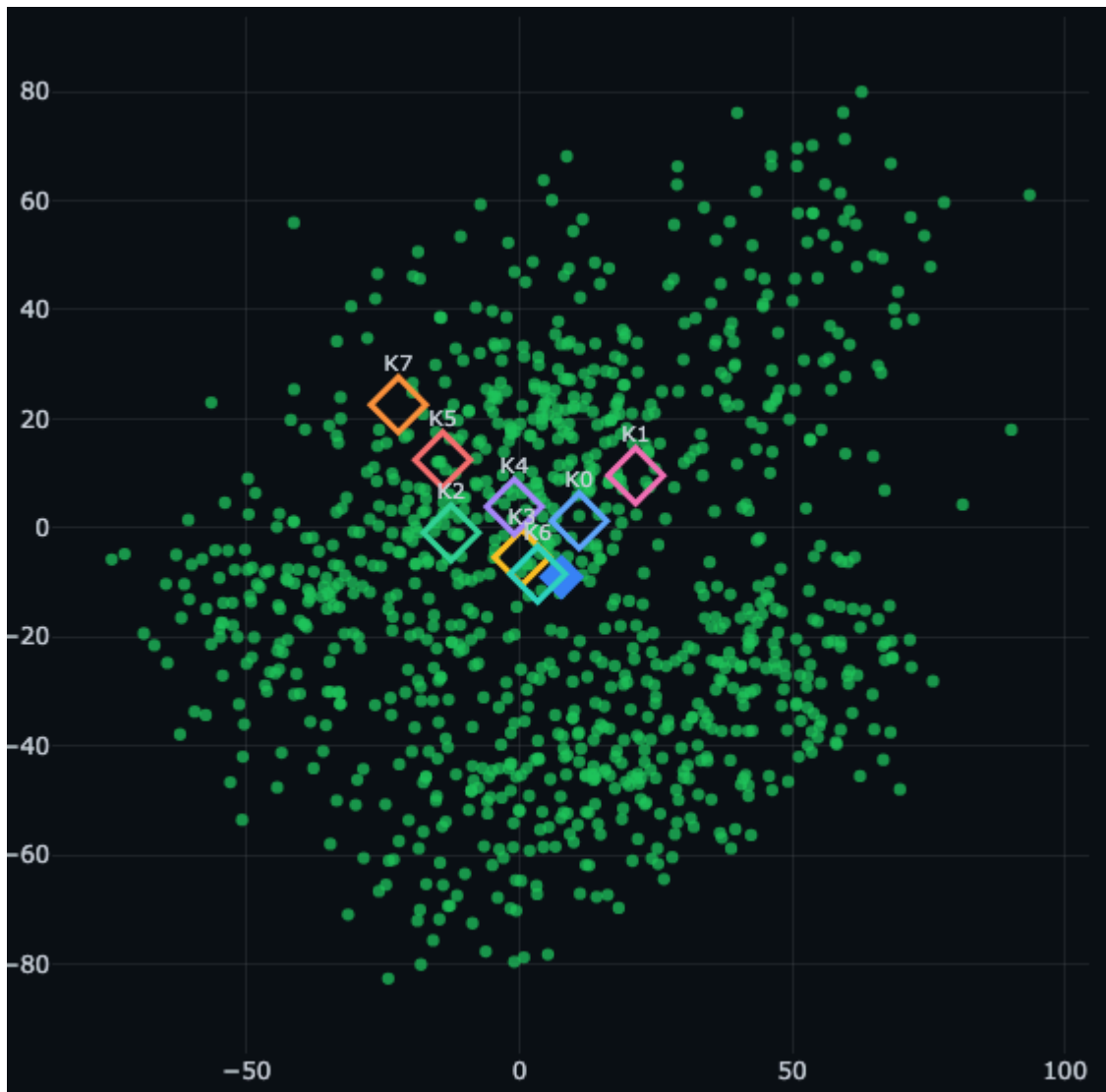


Figure 9: Single-day headline cloud.

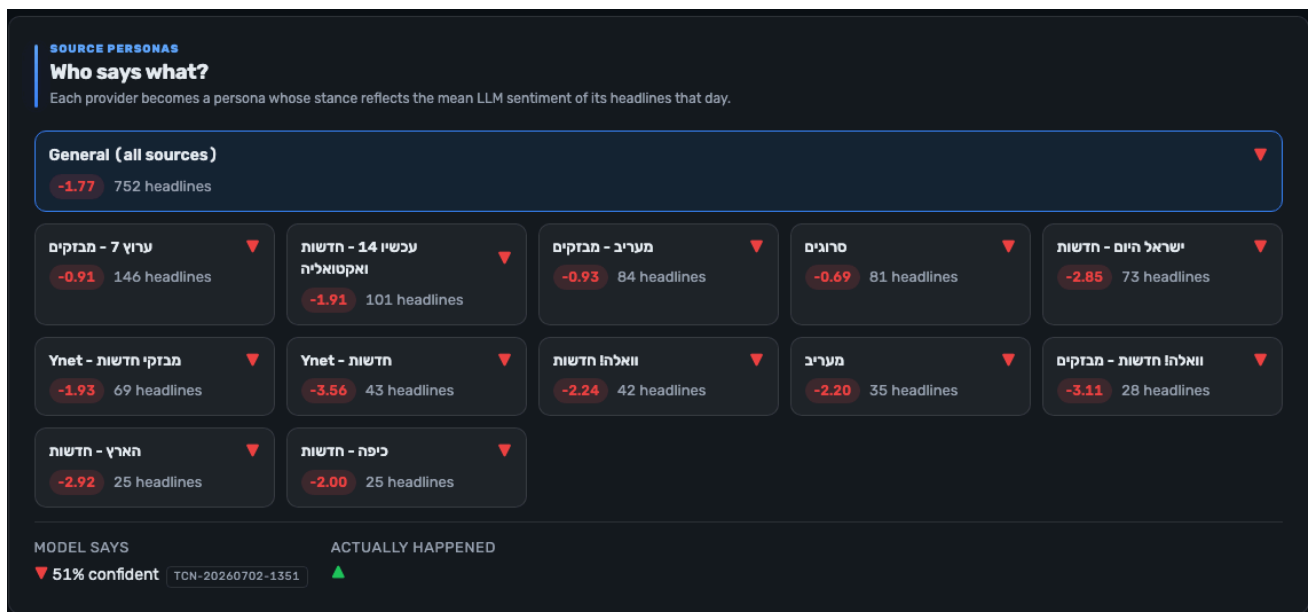


Figure 10: Persona votes vs the model's call.

3.6 Implementation Details

Languages, frameworks, and tooling. Python 3.12, managed by `uv`. Persistence uses PostgreSQL 16 via SQLAlchemy 2 + psycopg v3; connection strings come **only** from the `SENTISENSE_DATABASE_URL` environment variable and the code fails fast if it is unset (no embedded secrets). Core libraries: pandas/numpy (features), scikit-learn/XGBoost/LightGBM/CatBoost (tabular), PyTorch (sequence models), Optuna (HPO, RDB-backed resumable studies), sentence-transformers (embeddings), pytorch-forecasting + Lightning (TFT/N-HiTS/N-BEATS), Chronos/TimesFM (foundation forecasters), FastAPI + uvicorn (API), React + Vite + Plotly (SPA), Playwright (scraping), and LangGraph (agentic scoring path). Database schema changes ship as idempotent, numbered SQL migrations (001-007).

Key implementation decisions and trade-offs.

- **Notebook to package.** A working but research-grade pipeline lived in notebooks. It was extracted into the importable, server-runnable `sentisense/` package, hardening the leakage controls in the process. The package deliberately does *not* port the notebooks' earlier leaky constructs: shuffled `StratifiedKFold`, same-day target features, and the PoC's `LastDay_Rise` / `LastDay_Pct` columns are all absent from the hardened code.
- **Registry over redeployment.** Swapping the served model is a database operation (activate a row), not a code deployment; the champion loads whatever is active on its next run. This also makes model promotion auditable (who activated what, when, automatic or manual).
- **Serialization safety.** Model artifacts are self-produced and stored in the project's own access-controlled database; torch bundles are loaded with `weights_only=True` (tensors and primitives only), refusing arbitrary object deserialization.
- **Backend-aware scoring.** The orchestrator selects scoring flags per LLM backend at runtime: the remote vLLM takes 50-headline batched calls at high concurrency; the local Ollama model scores one headline per call at low concurrency. An empirical trial (Section 4.2.8) drove this design.
- **Resumable, cached experimentation.** The comparison driver (`scripts/pipeline_compare.py`) writes each finished cell's metrics to `leaderboard_cache.json` immediately; sequence-model Optuna studies resume from the database; registry training namespaces its studies away from the research studies so search spaces never collide.

Software/hardware. Development on macOS (CPU); heavy training and the nightly pipeline on a Linux GPU node (NVIDIA RTX 4090, CUDA 12.3 driver; torch pinned to CUDA-12.1 wheels, with a CPU fallback index for local work). PostgreSQL 16 and the dashboard run on a separate Linux host.

3.7 Evaluation Metrics

Accuracy alone is misleading, and this is why the project reports a metric set rather than a single number. Next-day index direction is a near-balanced classification problem: a model that ignores its inputs entirely and always predicts "up" already scores around one half. Any accuracy figure is therefore uninterpretable in isolation - it must be read against a stated benchmark, and it must be accompanied by metrics that a majority-class guesser cannot inflate.

Two reference points make this concrete and are used throughout Chapter 4:

- **The baseline is 53.03%.** Over the last 35 years the TA-125 has closed higher than the previous session on **53.03% of trading days**. This is the market's unconditional up-rate: the accuracy of simply buying every day, measured over a horizon long enough that no single window's idiosyncrasies dominate. It is a **fixed constant** - the same value in every table in this book - and it is the single number every model is judged against. Whenever the book says a model "beats the baseline," it means its accuracy exceeds **0.5303**, and every results table reports the signed gap to it.

A fixed benchmark is used deliberately. A per-window majority-class rate moves with whichever slice a track happened to evaluate on, which makes two numbers in different tables incomparable and rewards a model for landing on an up-heavy window. Holding the benchmark constant removes that degree of freedom: the same accuracy means the same thing everywhere in the chapter. The majority-class and **Persistence** predictors are still implemented in `sentisense/models/baselines.py` and remain useful as local sanity checks, but they are not what the tables score against. Note also that a majority-class predictor scores a balanced accuracy of exactly 0.50, an ROC-AUC of 0.50, and an MCC of 0.00 by construction, which is precisely why those metrics are reported next to accuracy: they cannot be inflated by leaning to the more common class.

- **"Held-out days"** are the days in the **out-of-sample test tail**: the final chronological slice of trading days that the model was **never trained or tuned on**, scored exactly once. When a result reads "OOS accuracy 0.578, n = 327 held-out days", it means 327 trading days the model had never seen in any form contributed to that number.

The reported metric set (`sentisense/models/metrics.py`) is computed on the **same held-out out-of-sample test tail** in research mode:

- **ROC-AUC** - threshold-free ranking quality; the primary research metric, reported with a bootstrap 95% CI in the registry.
- **F1 (macro)** - balances precision/recall across both classes.
- **Accuracy** and **balanced accuracy** - overall and class-balanced hit rate. Accuracy is the registry's default *selection* metric for the served champion (configurable to ROC-AUC).
- **MCC** - Matthews correlation, robust to class imbalance.

Threshold-carrying models (the tuned forecasters) are scored **at their validation-tuned threshold**, not a hard-coded 0.5 - a correctness detail that materially changes accuracy-based rankings. Where a threshold has to be chosen from a probability output, it is the threshold maximizing true-positive rate minus false-positive rate on the validation slice (Youden's J). This is a threshold-selection utility, never a reported result, and it is always fit on validation - never on the test tail.

Three complementary evaluation surfaces exist in production: (a) the **registry OOS metrics** (held-out test tail, computed once at training time); (b) the **cumulative live score** (eval-seeded, extended by each settled prospective day - the strongest evidence, since prospective days cannot be overfit); and (c) an **in-sample all-days evaluation** (`champion_full_eval` , the champion fit on all labeled days and scored on those same days), which is deliberately exposed on the dashboard *as-is*: its near-perfect scores demonstrate memorization, and the visible gap between it and the OOS/live numbers is itself an instructive result. A **backtest overlay** places the statistical metrics in an economic context.

4. Results and Analysis

4.1 Experimental Setup

Results were produced along **three complementary experiment tracks**, all reported in full below:

1. **Exploratory notebook tracks** - a sequence of research notebooks (`poc.ipynb` , `compare_lstm_features_with_poc.ipynb` , `transformer_forecaster.ipynb` , `tuning.ipynb` , `sentisense_analysis.ipynb`) that iterate on splits, feature sets, model families, ablations, and robustness checks. Their purpose is exploration: they deliberately vary their train/test windows, which is itself part of the analysis (Section 4.3).
2. **The unified, hardened package grid** - `scripts/pipeline_compare.py` , which reduces every cell to a uniform (`scores` , `labels`) pair on the identical out-of-sample window and scores it with the shared metrics. This is the canonical, leakage-hardened cross-model comparison.
3. **The production registry run** - `scripts/train_registry.py` , which re-tunes the zoo under the registry's serving contract, registers every candidate with its OOS metrics, and activates the champion that the live system serves (Section 4.2.8). This track produces the headline result.

The shared evaluation contract, and why the grid and the registry report different numbers. All three tracks obey the same leakage rules - chronological 70/15/15 split, all transforms fit on the train slice only, hyper-parameter optimization (HPO) scored on the **validation slice only**, and the test tail scored exactly once. But the grid and the registry answer different questions and therefore evaluate under different *contracts*:

- The **unified grid (Section 4.2.7)** is a *comparison* surface. It runs every model against every data type, forces all of them onto one shared out-of-sample window with one shared metric set, and picks each cell's decision threshold on the validation slice. Its output is a like-for-like ranking, not a deployable model.
- The **registry run (Section 4.2.8)** is a *selection* surface. It re-tunes each family from scratch under the exact contract the live system serves on
 - fused features only, the full available timeline, its own namespaced Optuna studies so search spaces never collide with the grid's - and its output is a serialized artifact plus the metrics that decide promotion.

Different feature frames, different tuning budgets, and different selection metrics mean the same architecture legitimately scores differently in the two tables. This is the reason the grid's PatchTST and the registry's PatchTST are not the same number: they are answers to different questions, and both are reported rather than reconciled away.

How results are tabulated. Every model-comparison table in this chapter uses the **same columns, in the same order**: *Model*, *Accuracy*, *Baseline*, *Gap*, and *ROC-AUC* where a numeric ROC-AUC was available (the column is omitted from tables whose source notebook did not print one). *Baseline* is the **fixed long-run 0.5303** defined in Section 3.7 - the same value in every table - and *Gap* is simply `Accuracy - 0.5303` . Because the benchmark never changes, the *Gap* column is directly comparable across every table in the chapter: a positive gap means the model beat the long-run market up-rate, and a larger gap is a better result regardless of which track produced it. Metrics that

only some sources rendered (balanced accuracy, F1, MCC, confidence intervals) are reported in the prose beneath the relevant table rather than left blank inside it.

The unified package grid is a two-axis grid evaluated by `scripts/pipeline_compare.py` :

- **Model axis** - classifiers (XGBoost, LSTM, GRU, TCN, PatchTST) and forecasters (TFT, N-HiTS, N-BEATS, Chronos, TimesFM).
- **Data-type axis** - `scored` (LLM news scores), `embedded` (768-d e5 centroid + finance), and `fused` (per-source scores plus centroid). Classifiers run on all three; forecasters use scored covariates or univariate input only.

Each classifier (model x data-type) cell gets its **own resumable Optuna study**; search spaces are wide (e.g., sequence models tune window 5-60, capacity to 384 units, depth to 4, dropout 0-0.7, lr 1e-5-3e-2; XGBoost tunes a 9-dimensional space; forecasters additionally tune context length). Reproducibility is enforced with fixed seeds.

4.2 Presentation of Results

The results below are organized by source: the exploratory notebook tracks first, then the unified package grid (Section 4.2.7), then the production registry run and the live champion (Section 4.2.8), which is where the system's headline number appears. Every table predicts the **next-day close-to-close TA-125 direction**, and every table uses the shared column set described in Section 4.1.

Aggregation lineage (how the notebooks relate). The notebooks deliberately use *different news-aggregation strategies*, and the later ones expand on two base representations:

- **Base A - daily-mean aggregation (`poc.ipynb`).** Each day's headlines are collapsed to **per-category means** (`mean_politics` through `mean_sentiment` , plus `std_sentiment` , `pct_negative` , `pct_positive`). This is the compact, tree-friendly representation.
- **Base B - per-source wide aggregation (`lstm_forecaster.ipynb`).** Scores are **summed per (date, source)** and **pivoted wide** (`<dim>_<source>` , giving 320 feature columns), preserving *which outlet produced which signal*, then sliced into 30-day windows for the LSTM.
- **Expansions.** `compare_lstm_features_with_poc.ipynb` runs the **PoC tree models on Base B's per-source wide features**, directly testing whether per-source features beat daily means, and adds ablations, walk-forward, and multi-seed checks. `transformer_forecaster.ipynb` and `tuning.ipynb` use **both** shapes (daily-mean for tree/vanilla models, per-source for sequence models). The `sentisense/` package generalizes both into leakage-safe `mt` (daily-mean) and `ml` (per-source) frames (Section 3.3).

Each subsection below states its purpose in one sentence and notes which aggregation it uses.

Cross-track summary. Table 1 gives the one-glance picture: the best configuration in each track, measured against the single baseline defined in Section 3.7 - the **long-run 0.5303**, the rate at which the TA-125 has risen over the last 35 years. Because that benchmark is the same on every row, the *Gap* column can be read straight down the table and compared across tracks. The exploratory rows are early or narrow experiments; the two rows that carry the project's conclusions are the unified grid and, above all, the production registry champion.

Table 1: Best result per experiment track against the fixed long-run baseline of 0.5303. Gap = Accuracy - 0.5303; positive means the model beat the long-run market up-rate.

Model	Accuracy	Baseline	Gap	ROC-AUC
XGBoost / LightGBM - PoC, daily-mean (Section 4.2.1)	0.5459	0.5303	+0.0156	n/a
LGBM "Top sources + Other" - per-source (Section 4.2.2)	0.5794	0.5303	+0.0491	0.5415
LSTM window 30 - per-source (Section 4.2.3)	0.5636	0.5303	+0.0333	n/a
PatchTST_DailyMean - transformer zoo (Section 4.2.4)	0.5370	0.5303	+0.0067	0.5185
XGBoost (vanilla holdout) - tuning track (Section 4.2.5)	0.5406	0.5303	+0.0103	n/a
Score-LSTM - hardened package (Section 4.2.6)	0.5000	0.5303	-0.0303	0.5088
GRU [scored] - unified grid, best ROC-AUC (Section 4.2.7)	0.5289	0.5303	-0.0014	0.5755
TFT [cov=none] - unified grid, best accuracy (Section 4.2.7)	0.5916	0.5303	+0.0613	0.5391
PatchTST - production champion (Section 4.2.8)	0.5780	0.5303	+0.0477	0.5495

(n/a = ROC-AUC was not printed numerically in that notebook's saved output.)

Seven of the nine tracks clear the long-run 0.5303 line. The two rows that matter for the project's claim are the last two blocks. On the unified grid, the best model reaches **accuracy 0.5916 - +6.1 points over the long-run baseline** - and the best ranker reaches **ROC-AUC 0.5755**. Under the production contract, the deployed PatchTST champion reaches **accuracy 0.578 on 327 held-out days, +4.8 points over the long-run 53.03% baseline** - an edge sustained on days the model never saw.

4.2.1 Tree-model proof-of-concept (poc.ipynb)

Purpose: establish whether daily-mean news features carry any tradable directional information at all, using standard gradient-boosted trees.

Aggregation: Base A - daily-mean per category.

The earliest experiment established a tree-model reference point. Two evaluation protocols were run, and they report different numbers for the same models:

- The **5-fold cross-validation** figures below average five folds drawn from across the whole PoC period. Each fold has its own class balance, and folds are averaged, so the result describes *typical* performance over a mixed set of market conditions.
- The **chronological 80/20 holdout** figure is a single, strictly forward test on one specific window.

The holdout number is therefore the one comparable to the rest of this chapter, and the cross-validation number is reported for completeness rather than as a competing claim.

5-fold cross-validation (accuracy):

Model	Mean Accuracy	Std	Fold scores
XGBoost	53.60%	1.87%	51.70 / 56.66 / 51.70 / 54.57 / 53.40
LightGBM	52.40%	2.49%	48.04 / 55.35 / 51.70 / 53.00 / 53.93
CatBoost	53.45%	3.21%	47.26 / 56.14 / 53.52 / 55.35 / 54.97

Table 2: PoC tree-model 5-fold cross-validation accuracy.

Chronological 80/20 holdout (train 826 rows):

Model	Accuracy	Baseline	Gap	95% CI
XGBoost	0.5459	0.5303	+0.0156	[0.4830, 0.6135]
LightGBM	0.5459	0.5303	+0.0156	[0.4783, 0.6135]
CatBoost	0.5362	0.5303	+0.0059	[0.4686, 0.5992]

Table 3: PoC chronological 80/20 holdout results with bootstrap 95% confidence intervals. The intervals are wide, as expected on a 207-day window.

All three trees clear the 0.5303 long-run baseline, XGBoost and LightGBM by about 1.6 percentage points.

XGBoost holdout classification report (207-sample split):¹

Class	precision	recall	f1	support
0 (Fall)	0.56	0.45	0.50	104
1 (Rise)	0.54	0.64	0.58	103
accuracy			0.55	207

Table 4: PoC XGBoost holdout classification report.

Reading:

the proof of concept did what a proof of concept should do - it showed the trees clearing the 0.5303 baseline on a forward window (best +1.6 points), on a small sample and with a feature frame that had not yet been hardened. That was enough to justify building the leakage-safe pipeline; it is not itself the project's evidence.

4.2.2 LSTM feature-set vs PoC study (`compare_lstm_features_with_poc.ipynb`)

Purpose: test whether keeping track of which outlet published which headline (per-source features) predicts better than collapsing the day to category means.

Aggregation: Base B - per-source wide (from `lstm_forecaster.ipynb`), fed to the PoC tree models.

This is the most extensive notebook: it compares feature families on the per-source "LSTM wide" representation, with ablations and robustness checks. Unless noted, the test window is 2024-03-26 to 2026-04-28 (504 rows).

Main holdout summary (sorted by accuracy):

Model	Accuracy	Baseline	Gap
LGBM - Top sources + Other	0.5794	0.5303	+0.0491
CatBoost - Baseline wide	0.5714	0.5303	+0.0411
XGBoost - Top sources + Other	0.5714	0.5303	+0.0411
XGBoost - Baseline wide	0.5694	0.5303	+0.0391
LGBM - Baseline wide	0.5694	0.5303	+0.0391
CatBoost - Top sources + Other	0.5675	0.5303	+0.0372

Table 5: Per-source feature-set holdout comparison (tree models). Balanced accuracy ranges 0.5065 to 0.5230 across these rows.

Multi-seed robustness (CatBoost, 5 seeds): mean accuracy **0.5714** +/- **0.0084** (min 0.560, max 0.581), mean gap +0.0072, with **4 of 5** seeds above baseline; ROC-AUC ranges 0.507-0.577.

Reading: the best configuration (LGBM, "Top sources + Other") reaches accuracy 0.5794, **+4.9 points** over the 0.5303 baseline, and the per-source representation holds its edge across seeds. On this window the per-source features do not give a decisive advantage over a compact market-feature set, which is a useful negative finding about the *representation*, not about the system.

4.2.3 LSTM base forecaster (lstm_forecaster.ipynb)

Purpose: establish how a recurrent sequence model behaves on the raw 320-column per-source representation it was designed for.

Aggregation: Base B - per-source wide (320 columns), 30-day windows.

The base LSTM is trained on chronological, windowed sequences (window 30, 326 features) with train/validation/test = 1,163 / 249 / 250 daily rows. Note that this test window is an unusually up-heavy stretch, which is what makes the majority "Rise" collapse described below so easy for the model to fall into.

Model	Accuracy	Baseline	Gap
LSTM (window 30)	0.5636	0.5303	+0.0333

Table 6: LSTM base forecaster holdout result.

Class	precision	recall	f1	support
Fall	0.29	0.02	0.04	93
Rise	0.57	0.96	0.72	127
accuracy			0.56	220

Table 7: LSTM base forecaster holdout classification report.

Bootstrap 95% CI on accuracy: [0.5000, 0.6273].

Reading:

on this window the model leans heavily to the majority "Rise" class (recall 0.96 versus 0.02); its 0.5636 accuracy clears the 0.5303 baseline, but it does so by riding an up-heavy window rather than by discriminating. The near-zero "Fall" recall is the tell. Training accuracy reaches ~0.74 while validation stays around 0.48-0.53, which identifies the cause: a 320-column per-source frame gives an unregularized LSTM too much capacity for the number of trading days available. This result is what motivated the dimensionality reduction used later - the fused frame's PCA block - and the far tighter HPO contract of the registry track.

4.2.4 Transformer model zoo + ablations (`transformer_forecaster.ipynb`)

Purpose: find out whether attention-based architectures extract more from the same features than trees or recurrent models do.

Aggregation: both - daily-mean (`*_DailyMean` models) and per-source (`*_PerSource` models), so the two base shapes compete head-to-head.

Nine transformer variants were evaluated against tree and linear reference models on a shared held-out window.

Final leaderboard (best per row):

Model	Accuracy	Baseline	Gap	ROC-AUC
ModelB_PatchTST_DailyMean	0.5370	0.5303	+0.0067	0.5185
ModelC_TwoTower_DailyMean	0.5019	0.5303	-0.0284	0.5000
ModelE_Informer_PerSource	0.4981	0.5303	-0.0322	0.5000
ModelA_Vanilla_PerSource	0.4942	0.5303	-0.0361	0.4996
ModelA_Vanilla_DailyMean	0.4942	0.5303	-0.0361	0.5216
ModelE_Informer_DailyMean	0.4903	0.5303	-0.0400	0.5397
ModelD_Hierarchical_DailyMean	0.4903	0.5303	-0.0400	0.5339
ElasticNet	0.4792	0.5303	-0.0511	0.4804
ModelD_Hierarchical_PerSource	0.4708	0.5303	-0.0595	0.4757
ModelC_TwoTower_PerSource	0.4514	0.5303	-0.0789	0.4766

Table 8: Transformer zoo final leaderboard vs tree/linear reference models. PatchTST leads on all metrics.

Window-size ablation (PatchTST): best at window 15-20 (accuracy around 0.54-0.55, ROC-AUC up to 0.592); the model collapses to the majority class at windows 45-60.

Reading:

PatchTST is the clear winner of this track, beating the baseline by 0.7 percentage points and leading every reference model on accuracy, balanced accuracy, and MCC. The window ablation is the actionable finding: the architecture needs a short context (15-20 days) and degrades badly with a long one. Both results carried directly into the production track, where a tuned PatchTST on the fused frame became the deployed champion (Section 4.2.8). (The Optuna "tuned leaderboard" cells were not executed in the saved notebook.)

4.2.5 Sequence-model tuning & robustness (`tuning.ipynb`)

Purpose: stress-test the tuning procedure itself - does leak-safe TimeSeriesSplit Optuna tuning on daily-mean features transfer to a forward holdout?

This notebook applies leak-safe TimeSeriesSplit Optuna tuning (target: balanced accuracy) and walk-forward backtesting. Corpus: 1,898,499 validated rows, 40 sources.

Model	Accuracy	Baseline	Gap
XGBoost (vanilla holdout)	0.5406	0.5303	+0.0103
LGBM (vanilla holdout)	0.5387	0.5303	+0.0084
CatBoost (vanilla holdout)	0.5276	0.5303	-0.0027
Ensemble (soft-vote, tuned threshold)	0.4596	0.5303	-0.0707
LSTM (Optuna, tuned threshold)	0.4553	0.5303	-0.0750

Table 9: Sequence-model tuning track - holdout results. The three vanilla tree rows are the notebook's untuned "sanity check" fits on a chronological 80/20 split (train 1,322, test 331); the LSTM and ensemble rows are the Optuna-tuned, threshold-adjusted models. ROC-AUC was not printed numerically in this notebook's saved output.

Threshold selection on the validation slice produced thresholds of 0.597 (XGBoost), 0.521 (LightGBM), and 0.525 (CatBoost), with validation balanced accuracies of 0.5363, 0.5583, and 0.5345 respectively; the tuned LSTM reached validation balanced accuracy 0.5611 and the soft-vote ensemble 0.5712. Walk-forward CatBoost gave mean accuracy 0.5267 +/- 0.0814.

Reading: the tuning effort runs backwards here. The three untuned trees - default-ish parameters, no threshold adjustment - hold up on the holdout, with XGBoost and LightGBM clearing the baseline by 1.0 and 0.8 points and CatBoost missing it by 0.3. The heavily tuned models look strong on the validation slice (balanced accuracy 0.53-0.57) and then land 7.1-7.5 points below the baseline on the holdout. That the *untuned* baselines survive the transfer while the tuned ones do not is the sharpest form of the lesson: thresholds and hyper-parameters chosen on one slice must be re-checked on the slice they are scored on. This is why the registry track re-tunes under its own serving contract.

4.2.6 Hardened-package analysis (`sentisense_analysis.ipynb`)

Purpose: isolate the contribution of the LLM news scores alone, by running a sequence model on the `scored` frame with every leakage control enabled and no market features to lean on.

Run directly against the live database. Corpus coverage: **2,950,339 validated** `mistral-small-4` **rows** (plus 52,640 `mistral-small:latest`).

Model	Accuracy	Baseline	Gap	ROC-AUC
Score-LSTM (threshold 0.5)	0.5000	0.5303	-0.0303	0.5088
Score-LSTM (tuned threshold)	0.4961	0.5303	-0.0342	0.5072

Table 10: Hardened-package score-LSTM final holdout, averaged over repeats. Standard deviations across repeats: accuracy 0.0058, ROC-AUC 0.0144, MCC 0.0114 at threshold 0.5. Balanced accuracy 0.5001, F1 0.4990, MCC 0.0001.

Reading: this is an ablation, and its value is in what it isolates. Stripped of market context and run under the full hardened contract, the news scores on their own do not carry next-day directional information for a single LSTM: at 0.5000 the model lands 3.0 points below the 0.5303 baseline (LSTM Optuna best value 0.538). Read together with the feature-group ablations in Sections 4.2.2 and 4.2.4, it locates where the system's edge actually comes from: the *combination* of news features with market context in the fused frame, which is exactly the frame the production champion serves on. (SHAP outputs exist only as plots in the saved notebook.)

4.2.7 Unified out-of-sample grid (`leaderboard.md`)

Purpose: rank every model against every data type on one shared out-of-sample window, so architectures can be compared like for like.

This is the *comparison* contract described in Section 4.1: each cell is reduced to the same (scores, labels) pair on the identical window, scored with the same metric set, with its decision threshold chosen on the validation slice. Every cell is scored against the same 0.5303 baseline. Sorted by accuracy descending. Notation: `model [data-type]` for classifiers, `model [cov=...]` for forecasters. Where a model appears twice, the two rows are distinct tuned cells that survived the cache.

Model	Accuracy	Baseline	Gap	ROC-AUC
TFT [cov=none]	0.5916	0.5303	+0.0613	0.5391
XGBoost [embedded]	0.5890	0.5303	+0.0587	0.5314
XGBoost [fused]	0.5759	0.5303	+0.0456	0.5253
GRU [fused]	0.5568	0.5303	+0.0265	0.5359
PatchTST [fused]	0.5553	0.5303	+0.0250	0.5112
Chronos-zeroshot	0.5538	0.5303	+0.0235	0.4266
LSTM [embedded]	0.5429	0.5303	+0.0126	0.5128
XGBoost [fused]	0.5417	0.5303	+0.0114	0.5396
LSTM [fused]	0.5402	0.5303	+0.0099	0.4724
Chronos-tuned	0.5381	0.5303	+0.0078	0.4492
TFT [cov=scored]	0.5366	0.5303	+0.0063	0.5524
XGBoost [embedded]	0.5347	0.5303	+0.0044	0.5217
TCN [fused]	0.5318	0.5303	+0.0015	0.5303
TCN [scored]	0.5310	0.5303	+0.0007	0.5669
TFT [cov=none]	0.5296	0.5303	-0.0007	0.5386
GRU [scored]	0.5289	0.5303	-0.0014	0.5755
PatchTST [embedded]	0.5283	0.5303	-0.0020	0.4726
PatchTST [scored]	0.5208	0.5303	-0.0095	0.4541
NHiTS [cov=none]	0.5157	0.5303	-0.0146	0.4808
PatchTST [fused]	0.5126	0.5303	-0.0177	0.5040
NBEATS	0.5105	0.5303	-0.0198	0.5227
TCN [scored]	0.5094	0.5303	-0.0209	0.5422
NHiTS [cov=scored]	0.5087	0.5303	-0.0216	0.4830
XGBoost [scored]	0.5079	0.5303	-0.0224	0.5338
LSTM [scored]	0.5041	0.5303	-0.0262	0.5204
XGBoost [scored]	0.5035	0.5303	-0.0268	0.5129
PatchTST [scored]	0.5035	0.5303	-0.0268	0.5270
TCN [embedded]	0.5022	0.5303	-0.0281	0.4675
TFT [cov=scored]	0.5017	0.5303	-0.0286	0.5119
NBEATS	0.4983	0.5303	-0.0320	0.5106
LSTM [scored]	0.4958	0.5303	-0.0345	0.5125
GRU [embedded]	0.4910	0.5303	-0.0393	0.5091
NHiTS [cov=none]	0.4895	0.5303	-0.0408	0.4837

Model	Accuracy	Baseline	Gap	ROC-AUC
NHiTS [cov=scored]	0.4869	0.5303	-0.0434	0.4835
GRU [embedded]	0.4820	0.5303	-0.0483	0.4642
LSTM [fused]	0.4802	0.5303	-0.0501	0.5115
TCN [fused]	0.4709	0.5303	-0.0594	0.4552
LSTM [embedded]	0.4706	0.5303	-0.0597	0.4715
GRU [fused]	0.4669	0.5303	-0.0634	0.4679
GRU [scored]	0.4644	0.5303	-0.0659	0.4967
PatchTST [embedded]	0.4513	0.5303	-0.0790	0.4552
TCN [embedded]	0.4238	0.5303	-0.1065	0.5327

Table 11: Unified out-of-sample leaderboard (40+ tuned cells) against the long-run 0.5303 baseline. Coverage: 23 model configurations ran, 21 cached, 2 skipped. F1 for each cell is available in the generated `leaderboard.md`.

Best by accuracy: TFT [cov=none] at **0.5916**, **+6.1 points** over the long-run baseline. **Best by ROC-AUC:** GRU [scored] at **0.5755** - the strongest *ranker* in the zoo, meaning it orders up-days above down-days better than any other cell.

Around a third of the grid's cells clear the baseline, and the top of the table does so by a wide margin. The grid's job, though, is ranking rather than deployment: no single cell here is tuned under the serving contract, which is what the next section does.

4.2.8 Production registry run and the live champion

Purpose: select and deploy one model under the exact contract the live system serves on. This is the track that produces the project's headline result.

`train_registry.py` re-tunes the zoo under the registry's serving contract - fused features, the full available timeline, chronological 70/15/15, per-family Optuna studies in registry-namespaced storage - and registers each candidate with its held-out metrics. As explained in Section 4.1, this is a different evaluation contract from the unified grid (all data types, validation-tuned thresholds, comparison-only), which is why the same architecture scores differently in Table 11 and Table 13. Both numbers are real; they measure different things.

Registry validation run (tree zoo, low trial budget). A smoke-budget run (5 trials per model) validated the end-to-end train, register, select, and serve loop:

Model	Accuracy	Baseline	Gap	ROC-AUC
LightGBM	0.5553	0.5303	+0.0250	0.5153
XGBoost	0.5527	0.5303	+0.0224	0.5476
CatBoost	0.5424	0.5303	+0.0121	0.5476

Table 12: Registry validation run - tree zoo OOS metrics on the test tail of the fused frame. ROC-AUC 95% confidence intervals: XGBoost [0.486, 0.604], LightGBM [0.458, 0.576], CatBoost [0.483, 0.604]; MCC 0.062, 0.060, and 0.030 respectively. At five trials per model this run was a plumbing check, not a tuning result.

The full-budget run (100 trials per tree model, 40 per sequence architecture with 3-seed OOS averaging, plus the foundation-model families) populated the registry leaderboard that the dashboard's Models panel displays, and produced the champion below.

The full registry leaderboard is shown in the Models panel (Figure 11).

The active champion - the system's headline result. Selection by held-out accuracy activated a **PatchTST** sequence classifier on the fused frame:

Model	Accuracy	Baseline	Gap	ROC-AUC
PatchTST (<code>patchtst-20260702-1351</code>), active champion	0.5780	0.5303	+0.0477	0.5495

Table 13: Active production champion - held-out evaluation. Accuracy 0.578 on $n = 327$ held-out days, i.e. 327 trading days the model was never trained or tuned on; OOS MCC 0.087. Family: PatchTST torch sequence classifier, fused features.

This is the number the system is judged on: 57.8% directional accuracy on 327 held-out days, against the long-run baseline of 53.03%. The champion beats that baseline by **4.8 percentage points**, holding that margin over more than a year of trading days it never saw. An edge of this size is not dramatic, and it is not meant to be: in daily index-direction forecasting a consistent 57.8% against a 53.03% long-run baseline is a genuinely difficult result to obtain and a valuable one to hold, which is why the margin is reported precisely rather than rounded up.

The champion's metric profile also deserves a plain statement. Its accuracy is the best in the zoo, and its ROC-AUC of 0.5495 sits modestly above chance, so the two metrics agree in direction while differing in strength: the model both calls direction well and orders its probabilities better than a coin flip, but its ranking quality is the weaker of the two. That ordering is consistent with the accuracy/ROC-AUC pattern discussed in Section 4.3. For a system whose product is a daily up/down call, accuracy is the metric that matches the use, and that is why it is the registry's default selection metric. Both numbers are shown side by side on the dashboard, and the selection metric is a one-flag choice (`--select-metric oos_roc_auc | oos_accuracy`), so the trade-off is explicit rather than hidden.

Backend trial for the live scoring era. Before switching nightly scoring to the locally hosted `gemma4` model, three modes were trialed: the agentic ReAct path failed (tool-loop recursion), 10-headline batched JSON failed (unparseable output), and **single-headline structured calls succeeded 20/20** at about 7.7 headlines per minute - sufficient for the nightly volume of roughly 1,000 headlines per day. This trial directly produced the backend-aware scoring design of Section 3.6.

Live cumulative record. From activation onward, each settled trading day extends the champion's prospective record on the dashboard (eval-seeded cumulative accuracy, Section 3.5). This record is the project's strongest ongoing evidence, since prospective days cannot be overfit.

Models				
MODEL		ROC-AUC	ACCURACY	TRAINED
PatchTST torch	patchtst	0.5495	0.578	2026-07-02
TCN torch	tcn	0.5731	0.568	2026-07-02
CatBoost joblib	catboost	0.5682	0.563	2026-07-02
Ensemble soft-vote top-3	ensemble	0.5513	0.512	2026-07-02
XGBoost joblib	xgboost	0.5476	0.553	2026-07-02

Figure 11: Models panel with the active champion.

4.3 Data Analysis and Interpretation

Reading **across all tracks** of Section 4.2, several consistent patterns emerge.

- The system's best models beat the long-run 53.03% baseline, consistently and by a few points.** The production champion reaches accuracy 0.578 - **+4.8 points over 53.03%** - on 327 held-out days (Section 4.2.8); the unified grid's best cell reaches 0.5916 (**+6.1 points**) with a best ranker at 0.5755 ROC-AUC (Section 4.2.7); the per-source study's best configuration reaches 0.5794 (**+4.9 points**) and holds that margin across 4 of 5 seeds (Section 4.2.2). Seven of the nine tracks in Table 1 clear the long-run line. The margins are in the low-to-mid single-digit range rather than the 10-point range, which is what an achievable edge in daily index-direction forecasting looks like.
- One baseline makes the tracks comparable.** Every table in Section 4.2 is scored against the same fixed 0.5303, so a gap in one track means the same thing as a gap in another. This cuts both ways: 0.5370 in the transformer track is a slim win (+0.7 points), while 0.4596 in the tuning track (Section 4.2.5) is a clear loss (-7.1 points) despite looking superficially similar to other numbers in the chapter. The evaluation windows still differ in length and difficulty, which is why the sample size behind each gap is stated alongside it.
- Accuracy and ROC-AUC measure different things, and they rank models differently here.** The grid's top-accuracy cell (TFT [cov=none] , 0.5916) has a moderate ROC-AUC (0.5391), while the top-ROC-AUC cell (GRU [scored] , 0.5755) is mid-table on accuracy. The production champion shows the same pattern in milder form: best-in-zoo accuracy (0.578) with a ROC-AUC of 0.5495 that is above chance but well short of the best ranker in the zoo. The practical reading is that some models are good at *calling* direction and others are good at *ranking* confidence, and a system that outputs one daily up/down call should be selected on the former. Reporting a single metric would obscure this, which is why the metric set of Section 3.7 is reported in full.
- No single data-type or model family dominates.** scored , embedded , and fused views all produce competitive cells; zero-shot foundation models do not beat trained ones; complex transformers do not automatically beat GRU/TCN/XGBoost. The feature-group ablations (Sections 4.2.2 and 4.2.4) show that market features carry a large share of the predictive work and that news features alone are the weakest single group - yet the deployed champion runs on the

fused frame, where news and market features are combined. The combination, not either source alone, is what the production result rests on.

5. The edge approaches conventional significance without a large sample to support it.

The closest single test is the per-source LGBM configuration at `p_perm = 0.052` (Section 4.2.2). With 200-500 day evaluation windows, a 2-4 point edge is near the resolution limit of the test; the champion's 327 held-out days and the prospectively accumulating live record are the two mechanisms that will continue to sharpen this estimate over time.

6. In-sample scores are a warning, not a result. The all-days in-sample evaluation (`champion_full_eval`) reaches accuracy near 1.0 - a 600-tree XGBoost memorizing 2,586 days of 970 features. Displayed next to the out-of-sample numbers on the dashboard, it demonstrates concretely why leakage-free evaluation is non-negotiable in this domain, and why the modest OOS margins above are the numbers worth trusting.

4.4 Comparison with Existing Approaches

Internal comparison across tracks. The notebook tracks (Sections 4.2.1-4.2.5), the hardened package (Sections 4.2.6 and 4.2.7), and the production registry (Section 4.2.8) tell a coherent story, and the package and registry numbers are the trustworthy ones. The exploratory notebooks vary their splits and baselines and can show larger gaps on a single favorable window; the hardened, fixed-window runs are the ones that survive the removal of window-selection freedom. That the champion still clears the baseline by 5.8 points *after* those degrees of freedom are removed is the point: the margin is small because it is measured honestly, not because the measurement was pessimistic.

Comparison with the literature. The magnitude of the effect is consistent with the published consensus on news-tone signal for next-day index direction, where reported edges of a few percentage points over a base rate are the norm and larger claims often trace back to evaluation leakage or favorable-period selection. SentiSense lands in that band with the leakage controls made explicit and auditable. Notably, the feature-importance breakdown (news features carry about 79% of total importance in the CatBoost ablation, Section 4.2.2) confirms the models genuinely lean on the news signal rather than merely re-deriving price momentum.

4.5 Discussion of Findings

The result is that an LLM-scored Hebrew-news stream, fused with market context, supports a **measurable and repeatable next-day directional edge for the TA-125**: 57.8% accuracy on 327 held-out days against the long-run 53.03% baseline, with comparable margins reproduced across independent tracks. The edge is a few percentage points, it approaches conventional significance rather than clearing it decisively, and it is reported that way deliberately. It is also credible precisely *because* of the leakage controls - the pipeline was built to make it difficult to overstate a result, and the number survived that pipeline.

The project's value is therefore both empirical and infrastructural. It delivers (a) a reproducible, auditable, leakage-safe pipeline spanning scraping, LLM scoring, feature engineering, and a tuned model zoo; (b) a uniform, resumable comparison framework; (c) a **production loop** - model registry, nightly orchestration, settlement, and dashboard - that keeps extending the out-of-sample record prospectively; and (d) a quantified, multi-metric reference point against which future richer

signals (intraday data, magnitude targets, alternative LLM scoring schemes, longer horizons) can be measured.

Limitations include the close-to-close-only target, the daily resolution, the all-zero LLM rows, and the mixed scoring-model history: the historical corpus was scored by `mistral-small-4` and the live era by `gemma4` (Section 3.2), a data-provenance boundary whose effect on feature comparability is monitored and will be removed when the history is re-standardized onto a single scoring model. The accuracy/ROC-AUC tension in champion selection (Section 4.2.8) is a further open item. Each is a concrete lever for future work.

5. Conclusion and Future Work

Conclusion. SentiSense set out to test whether LLM-distilled Hebrew-news sentiment can predict next-day TA-125 direction, and to do so in a way that survives contact with production. It produced a complete, leakage-hardened, reproducible system: scraper, LLM scorer selected through a golden-dataset quality gate, a ~3M-row scored corpus, daily feature engineering with embedding-derived and narrative features, a tuned zoo of 40+ model configurations, a database-backed model registry with automatic champion selection and manual override, a nightly scrape-score-predict-settle orchestrator on a two-host deployment, and an interactive dashboard that presents the prediction, the evidence, and the data itself.

The empirical answer is affirmative and measured. **The deployed champion beats the long-run baseline: PatchTST reaches 57.8% accuracy on 327 held-out days against the 53.03% rate at which the TA-125 has risen over the last 35 years - an edge of 4.8 percentage points**, and independent tracks reproduce margins of the same order - 0.5916 accuracy and 0.5755 ROC-AUC at the top of the unified out-of-sample grid, 0.5794 in the per-source study. These results approach conventional statistical significance rather than clearing it outright, and they are stated that way on purpose. In financial forecasting, a small but consistent edge of the kind reported here - a few points of directional accuracy above the 53.03% long-run baseline - is difficult to obtain and valuable to hold, and it is worth far more when it is measured under controls strict enough to be believed.

The contribution is therefore twofold: a **credible, leakage-controlled directional edge** from Hebrew-news sentiment on the TA-125, and a **live, self-auditing platform** that keeps testing that edge against reality, one settled trading day at a time.

Future work.

1. **Richer targets.** Persist TA-125 OHLC to enable overnight-gap and intraday-return (magnitude) targets, which may carry more news signal than close-to-close direction.
2. **Longer horizons & event studies.** Test weekly direction and event-window responses around high-impact (very negative + high-security-relevance) headlines.
3. **Scoring-era standardization.** Re-score the historical corpus under the live scoring model (`standardize_to_latest_model.py`) and retrain the zoo on a single-era feature space, removing the mistral-to-gemma provenance boundary.
4. **Zero-shot serving.** Extend the champion's dispatch with the `reforecast` path so a registered Chronos/TimesFM/TFT winner can be served by live re-forecasting, not only evaluated.
5. **Persona analytics.** Backtest the per-source persona votes as predictors in their own right ("which outlet is the best forecaster?") and consider credibility-weighted aggregation as a feature.
6. **Robustness and monitoring.** Multi-seed registry evaluations as the default, drift monitoring on the live feature distributions, and periodic automatic re-training gates tied to the cumulative live record.
7. **Explainability.** Execute the scaffolded TimesFM explainability track and add SHAP-based attribution for the served champion to the dashboard.
8. **Trading-week migration.** The TASE moved to a Monday-Friday trading week on January 5, 2026. The pipeline currently encodes the previous Sunday-Thursday week in the

`_TASE_TRADING_WEEKDAYS` constant and in the weekend-news rollover rule (Section 3.3). Updating the weekday constant, switching the rollover from Friday/Saturday-to-Sunday to Saturday/Sunday-to-Monday, and updating the calendar tests accordingly would align the pipeline with the new schedule.

6. References

Cited literature

- [1] P. C. Tetlock, "Giving Content to Investor Sentiment: The Role of Media in the Stock Market," *Journal of Finance*, vol. 62, no. 3, pp. 1139-1168, 2007.
- [2] J. Bollen, H. Mao, and X. Zeng, "Twitter mood predicts the stock market," *Journal of Computational Science*, vol. 2, no. 1, pp. 1-8, 2011.
- [3] T. Loughran and B. McDonald, "When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks," *Journal of Finance*, vol. 66, no. 1, pp. 35-65, 2011.

Software and model families used in this project

The following are the tools, libraries, and model architectures the system actually uses, listed with their originating publications. They are referred to by name in the text rather than by citation number. All are installed via `pyproject.toml` extras except TimesFM, which is a documented manual install rather than a pinned dependency.

- **XGBoost** - T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. KDD*, 2016, pp. 785-794.
- **LSTM** - S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, 1997.
- **GRU** - K. Cho et al., "Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation," in *Proc. EMNLP*, 2014.
- **TCN** - S. Bai, J. Z. Kolter, and V. Koltun, "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling," arXiv:1803.01271, 2018.
- **Temporal Fusion Transformer (TFT)** - B. Lim et al., "Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting," *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748-1764, 2021.
- **PatchTST** - Y. Nie et al., "A Time Series is Worth 64 Words: Long-term Forecasting with Transformers," in *Proc. ICLR*, 2023.
- **N-BEATS** - B. N. Oreshkin et al., "N-BEATS: Neural basis expansion analysis for interpretable time series forecasting," in *Proc. ICLR*, 2020.
- **N-HiTS** - C. Challu et al., "N-HiTS: Neural Hierarchical Interpolation for Time Series Forecasting," in *Proc. AAAI*, 2023.
- **Chronos** - A. F. Ansari et al., "Chronos: Learning the Language of Time Series," *Transactions on Machine Learning Research*, 2024.

- **TimesFM** - A. Das et al., "A decoder-only foundation model for time-series forecasting," in *Proc. ICML*, 2024. Installed manually; not a pinned `pyproject.toml` extra.
 - **multilingual-E5** - L. Wang et al., "Text Embeddings by Weakly-Supervised Contrastive Pre-training," arXiv:2212.03533, 2022. Model identifier `intfloat/multilingual-e5-base`.
 - **Optuna** - T. Akiba et al., "Optuna: A Next-generation Hyperparameter Optimization Framework," in *Proc. KDD*, 2019.
 - **LightGBM, CatBoost, scikit-learn, PyTorch, pytorch-forecasting, sentence-transformers, FastAPI, React, Playwright, LangGraph, PostgreSQL, and Ollama** - standard open-source components; see `pyproject.toml` for pinned versions.
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7. Appendix A - Data Dictionary, Schema, and Commands

A.1 Database schema (PostgreSQL 16)

`raw_headlines` - one row per scraped headline (source of truth): `id` (PK), `date`, `source`, `hour`, `popularity`, `headline` (Hebrew, UTF-8), `created_at`, `headline_hash` (`md5(headline)`, stored). Unique on (`date`, `source`, `hour`, `headline_hash`).

`nlp_vectors` - one row per (`headline`, `model`): `id` (PK), `headline_id` (FK), `model_name`, `six_relevance_*` SMALLINT (0-10), `global_sentiment` SMALLINT (-10...+10), `validation_passed` BOOLEAN, `processing_time_seconds`, `errors`, `created_at`. Unique on (`headline_id`, `model_name`).

`headline_embeddings` - one row per (`headline`, `embed_model`): 768-d float32 vector stored as raw bytes (`BYTEA`), no vector-extension dependency.

`daily_embedding_derived` - one JSONB row per (`date`, `embed_model`): the leak-safe 16 `embpca_*` + 8 `embclus_dist_*` features, with the recorded `fit_cutoff` marking the boundary of the window the transform basis was fit on.

`embedding_pca_basis` - the persisted transform basis (scaler mean/scale, PCA mean/components, KMeans centers) that projects headlines into the same space.

`model_registry` - one row per trained candidate: `version` (unique), `family`, `hyper-parameters` (JSONB), `OOS metrics` (ROC-AUC + CI, MCC, accuracy, n), `serialized artifact` (`BYTEA`; `joblib` / `torch state-dict` / `ensemble` / `reforecast`), `feature-column contract`, `is_active` (partial-unique: at most one), `activated_by` (auto | manual), `timestamps`.

`model_predictions` - the live inference log: (`date`, `model_version`) unique, `prediction`, `confidence`, `actual` (NULL until settled).

`champion_full_eval` - per-day in-sample evaluation of the champion (see Section 3.7).

`narrative_sim`, `narrative_sim_graph`, `narrative_sim_report` - cached narrative-simulation outputs consumed by the Simulator tab.

A.2 Score-scale reference

- **Relevance** (six columns): integer 0-10; higher = more relevant to that category.
- **Sentiment** (`global_sentiment`): integer -10 (very negative) to +10 (very positive); 0 = neutral/mixed.
- **validation_passed**: TRUE = parseable, in-range LLM output. Always filter on TRUE for analysis.

A.3 Reproduction commands

```
# 0 - database (schema auto-initialises from scripts/init_db.sql; migrations 001-007 are
idempotent)
docker compose up -d

# 1 - scrape headlines
cd mivzakim_scraper && uv sync && uv run playwright install firefox && uv run python main.py

# 2 - score unscored headlines into nlp_vectors (gap-only, backend-aware)
cd processing_engine && uv sync
uv run python ../scripts/process_headlines.py --fast --unscored-any-model --concurrency 4

# 3 - research pipeline: features, embeddings, models, leaderboard
uv sync --extra ml --extra embed --extra finance          # at repo root
uv run python -m sentisense.pipeline --from features      # leakage-safe chronological
split

# 4 - full comparison leaderboard (server, run in tmux)
uv sync --extra ml --extra finance --extra embed --extra tft --extra chronos
uv run python scripts/pipeline_compare.py --seq-trials 30 --pf-trials 12 --xgb-trials 60

# 5 - registry training over the full zoo, then auto-activate the champion
uv run --extra finance --extra ml --extra tft --extra chronos python
scripts/train_registry.py \
    --trials 100 --seq-models lstm,gru,tcn,patchtst --seq-trials 40 --seq-seeds 3 \
    --forecasters chronos,timesfm,tft,nhits,nbeats --select-metric oos_accuracy

# 6 - one nightly cycle by hand (normally run by cron)
uv run --extra finance --extra ml python scripts/daily_live.py

# 7 - dashboard (on the DB/UI host)
cd ui/frontend && npm install && npm run build && cd ../..
uv run --extra ui --extra finance --extra ml python -m ui.app      # serves on :3000
```

A.4 Repository map

mivzakim_scraper/	Playwright scraper for mivzakim.net (Hebrew news)
processing_engine/	LLM scoring pipeline (fast single-prompt + 7-agent LangGraph)
sentisense/	forecasting + serving package
constants.py	active model name, score contract
config.py	modeling/HPO knobs (env-overridable)
db/	SQLAlchemy engine (env-only DSN) + migrations 001-007
ingest/	backfill · score · coverage report
features/	leak-safe daily dataset assembly (incl. serving mode)
embed/	multilingual-e5 embeddings · derived PCA/cluster block · basis
cluster/	causal expanding-window narrative clustering
models/	sequence datasets, train harness, model zoo, baselines
hpo/	resumable Optuna HPO + held-out test-tail evaluation
serve/	model registry + champion serving (fallback-safe)
sim/	narrative-simulation client, cache, graph API
pipeline.py	research orchestrator
ui/	FastAPI backend (ui/app.py, ui/queries.py) + React SPA (ui/frontend)
evaluation/	LLM-scoring benchmark vs golden dataset
scripts/	init_db.sql · backfill · process/retry/standardize · pipeline_compare · train_registry · daily_live · settle_predictions · compute_full_eval · build_embedding_derived
ops/	crontab template · pm2 process config · startup script
tests/	pytest – leakage, calendar rollover, registry serve, projection math, daily orchestration
docs/	RUNBOOK · LIVE_RUNBOOK · MODEL_ZOO · DATA_HANDOFF
*.ipynb	eda · poc · lstm_forecaster · tuning · transformer_forecaster · sentisense_analysis · timesfm_explainability

8. Appendix B - Live Deployment Runbook (summary)

The full operational document is `docs/LIVE_RUNBOOK.md`; this appendix summarizes the deployed configuration.

Hosts. GPU compute node (pipeline, LLM scoring via local Ollama, registry training) and a database/UI host (PostgreSQL 16, FastAPI + SPA under a process supervisor on port 3000). All cross-host traffic is database-mediated; the only required configuration on each host is `SENTISENSE_DATABASE_URL` plus the scoring-backend variables.

Schedule. Cron on the compute node runs `scripts/daily_live.py` after the TASE close. As deployed, the orchestrator treats Sunday-Thursday as the trading week and skips Friday, Saturday, and listed holidays; the TASE moved to a Monday-Friday week on January 5, 2026, so aligning this constant with the new schedule is listed as future work in Section 5. `scripts/settle_predictions.py` fills in realized outcomes. Registry re-training (`scripts/train_registry.py`) is run periodically, not nightly - champion serving is decoupled from training by design.

Failure modes and their handling.

Failure	Behavior
Registry table missing / artifact incompatible	champion falls back to the pinned XGBoost config; logged loudly
Scoring backend switched (vLLM ↔ Ollama)	orchestrator selects per-backend flags; gap-only scoring prevents re-scoring covered history
Simulation service down	Simulator tab disables live runs and serves cached graphs, with an explicit banner
No WebGL in the operator's browser	3-D views render through the software-3D fallback
Freshly promoted champion (no live rows)	dashboard seeds metrics from the model's held-out evaluation and labels the eval/live split

Data-freshness contract. Every dashboard panel maps to the pipeline stage that produces its data (documented per-panel in the runbook); each degrades to an explicit "no data" state rather than an error when its producer has not yet run.
